2019-Summer Paper 2 – Urology

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Q1. Describe in detail the management of T1G3 TCC of urinary bladder?

Type: LAQ (25 marks)

Introduction

- T1G3 is high-grade (HG) urothelial carcinoma (the current name for TCC, transitional cell carcinoma) that invades the lamina propria but not the muscularis propria.
- T1 is 20–30% of NMIBC (non-muscle-invasive bladder cancer) and its most dangerous form: recurrence >80%, progression to muscle invasion 30–50%.
- Grade, not stage, drives progression. So every HG T1 is **high risk** (AUA [American Urological Association]/SUO [Society of Urologic Oncology] and EAU [European Association of Urology]).

Risk factors for progression

Factor	Significance
Concurrent CIS (carcinoma in situ)	Second most important factor after grade
Depth of invasion (substaging)	Invasion deep to the muscularis mucosae is worse; T1a/T1b proposed, no agreed method
LVI (lymphovascular invasion)	Suggests occult spread
Variant histology	Micropapillary (any amount), plasmacytoid, sarcomatoid, nested
Size >3 cm, multifocal, recurrent	More residual tumour and progression
Prostatic urethra, diverticulum	Poor BCG contact; diverticulum lacks muscle
Residual T1 at re-TURBT	Progression up to 80%
HG T1 at 3 months after BCG	Progression about 82%

- **Also know:** EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables; hydronephrosis suggests muscle invasion.

Evaluation

- Cystoscopy, urine cytology, CT urography (upper tract tumour <5%), bimanual examination under anaesthesia.

Step 1: Complete, good-quality TURBT (transurethral resection of bladder tumour)

- Resect all visible tumour **with detrusor muscle** in the specimen; send the base separately.
- Blue-light cystoscopy or NBI (narrow-band imaging) finds more CIS. **En bloc resection** keeps orientation for substaging.
- Biopsy the prostatic urethra if tumour is at the bladder neck or trigone, or with CIS.

Step 2: Re-TURBT (restaging) in all T1

- Within **2–6 weeks**, before BCG (bacille Calmette-Guérin).
- **Why:** residual tumour in ≥40%; upstaging to muscle invasion in 25–30% (about half if no muscle in the first specimen). Divrik's RCT (randomised controlled trial) showed better recurrence-free and progression-free survival.

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HG T1 on TURBT → re-TURBT at 2–6 weeks
   ├─ ≥T2                          → neoadjuvant chemotherapy + cystectomy
   ├─ residual T1 / very high risk → offer early radical cystectomy
   └─ T0, Ta or CIS                → BCG induction + 3-year maintenance
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Step 3a: BCG – standard for most patients

- **Mechanism:** BCG attaches to fibronectin, triggers cytokines (IFN-γ [interferon-gamma], IL [interleukin]-2) and a **Th1 (T-helper 1)** response that kills tumour. It is the **only intravesical agent that reduces progression**.
- Start 2–4 weeks after TURBT; induction weekly ×6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** 3 weekly doses at 3 and 6 months, then every 6 months to **3 years**. Median recurrence-free survival 76.8 vs 35.7 months. **EORTC 30962:** full dose for 3 years in high risk.
- **Contraindications:** gross haematuria, traumatic catheterisation, symptomatic UTI (urinary tract infection), previous BCG sepsis.
- **Toxicity:** cystitis common; BCG sepsis needs isoniazid, rifampicin, ethambutol and steroids.

Step 3b: Early (immediate) radical cystectomy

- **Rationale:** clinical T1 is understaged in one-third to one-half. Patients who progress under surveillance do worse than those presenting with muscle invasion. Immediate cystectomy gives about 92% 10-year disease-free survival.
- **AUA:** offer to fit patients with persistent HG T1 on re-resection, or T1 with CIS, LVI or variant histology.
- **EAU very-high-risk group (2021 model; risk factors: age >70, multiple papillary tumours, diameter ≥3 cm):** T1 HG with CIS plus ≥1 risk factor; T1 HG without CIS but with all three; also LVI, variant histology, CIS of the prostatic urethra. Offer immediate cystectomy; BCG if declined or unfit.
- Also: diverticular tumour, tumour not controllable endoscopically.

Step 4: BCG failure

- **Adequate BCG for T1:** ≥5 of 6 induction doses.
- **BCG-unresponsive:** HG T1 at the 3-month check after induction; persistent or recurrent CIS or HG Ta/T1 despite adequate BCG; or HG relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.
- **Standard:** timely radical cystectomy (EAU, AUA).

- **Unfit or refusing:** pembrolizumab (KEYNOTE-057: CIS complete response 41%); nadofaragene firadenovec (53%); N-803 with BCG; gemcitabine–docetaxel; TAR-200; trials.
- **Also know:** BCG-intolerant; delayed relapse may respond to BCG rechallenge.

Follow-up (AUA/SUO high risk)

- Cystoscopy and cytology every 3–4 months for 2 years, 6-monthly for 2 years, then yearly for life.
- Upper-tract imaging yearly for 2 years.
- Positive cytology with normal cystoscopy: check prostatic urethra and upper tracts.

Recent advances

- Checkpoint inhibitors with BCG in BCG-naive disease (trials); urine markers predicting BCG response.

Diagram to draw: bladder wall layers showing Ta, T1a/T1b (above and below the muscularis mucosae) and T2.

> **Key points to remember:**
> - Re-TURBT in all T1 at 2–6 weeks; residual T1 means progression up to 80%.
> - BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
> - Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy.
> - BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.

Q2. Describe indication, mechanism of stone fragmentation and current status of ESWL?

Type: LAQ (25 marks)

Introduction

- ESWL (extracorporeal shock wave lithotripsy), now SWL, breaks stones with focused acoustic shock waves generated outside the body. First human use 1980 (Dornier HM1).
- A shock wave is a short pulse: peak positive pressure 30–100 MPa, then a negative tail of about −10 MPa.

Components of a lithotripter

- Energy source at F1 (first focus); focusing system; coupling (water or gel); fluoroscopy and/or ultrasound to place the stone at F2. A focal zone wider than the stone improves breakage.

Energy source	Principle	Advantages	Disadvantages
Electrohydraulic (HM3)	Underwater spark; ellipsoid reflector	Most effective	Electrode wear; variable shocks
Piezoelectric	Ceramic elements on a spherical dish	Long life; little pain	Low energy; tiny focal zone
Electromagnetic (modern machines)	Coil moves a metal membrane; lens focuses	Consistent; high energy at F2	—

Mechanisms of stone fragmentation

- Each shock creates stress inside the stone. Above a critical value, cracks form; repeated shocks make them grow and join (**fatigue**).

Mechanism	How it works
Spallation	Wave reflects and inverts at the back surface of the stone; tension breaks off the distal face
Shear and tear (superfocusing)	Waves inside the stone reflect and focus; cracks form where shear peaks
Quasistatic squeezing	Wave travels faster in stone than fluid; a ring of pressure compresses the stone
Dynamic squeezing	Shear waves at the stone edges combine; strongest force in initial breakage
Cavitation	Negative phase forms bubbles; collapse erodes the surface. Excess bubbles shield the stone and injure tissue

Indications

- **Renal (AUA/EAU):** <2 cm outside the lower pole and lower pole <1 cm (SWL or URS [ureteroscopy]); EAU: 10–20 mm equal to endourology; >20 mm only after PCNL (percutaneous nephrolithotomy).
- **Ureteric:** any level, best <10 mm; emergency SWL within 48 hours works well. Patients avoiding anaesthesia and stents; children.
- **Poor choices:** >2 cm or staghorn; lower pole >1 cm (**Lower Pole I**: stone-free 21% SWL vs 91% PCNL); cystine, brushite, calcium oxalate monohydrate; infection stones.

Contraindications

- **Absolute:** pregnancy; uncorrected coagulopathy or antithrombotics; untreated UTI (urinary tract infection); obstruction distal to the stone; nearby arterial aneurysm (aortic >4 cm); severe uncontrolled hypertension.
- **Relative:** obesity or skeletal deformity preventing targeting; pacemaker.

Predictors of success

- **Stone:** size (most important); density <900 HU (Hounsfield units) clears best, >1000 HU predicts failure; composition; impaction.
- **Patient:** SSD (skin-to-stone distance) >10 cm predicts failure.
- **Lower pole anatomy:** steep infundibulopelvic angle, long (>10 mm) or narrow (<5 mm) infundibulum.
- **Triple D score:** density, volume and skin-to-stone distance.

Technique to optimise results

1. Treat infection; stop antithrombotics; control blood pressure. No routine stent: it does not improve clearance.
2. Good analgesia or sedation, because stone movement reduces breakage.
3. **Coupling:** gel as a mound, free of air; air pockets cut fragmentation by up to two-fifths.
4. **Rate 60–90 shocks/min** beats 120/min: bubbles clear between shocks, so less shielding and renal injury.
5. **Ramping:** start at low energy and step up; this causes protective vasoconstriction.
6. Monitor with imaging; more than two sessions add little.
7. **Adjuncts:** α-blockers; percussion and inversion for lower pole fragments.

Complications

- Renal haematoma (risk: age, hypertension, obesity, diabetes, high energy, many shocks); steinstrasse (risk rises with burden >2 cm^2); colic, sepsis; arrhythmia (ECG [electrocardiogram] gating); rare liver, spleen or bowel injury.
- Long-term hypertension, diabetes and infertility are **not** proven.

Current status

- **Declining use worldwide.** URS has grown with small digital scopes and lasers, giving higher single-session stone-free rates. Newer lithotripters are also less effective than the HM3.
- **SWL vs URS:** SWL has fewer complications, no stent or general anaesthesia, lower cost, but more retreatment. Guidelines still list it first-line for small renal and ureteric stones.

Recent advances

- **Burst wave lithotripsy:** bursts of 5–100 ultrasound cycles at 4–8 MPa; less cavitation; done awake; uniform fragments.
- **Ultrasonic propulsion** moves fragments to help them pass.

Diagram to draw: ellipsoid reflector with F1 (spark) and F2 (stone), with spallation and squeezing arrows on the stone.

> **Key points to remember:**
>
> - Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
> - Failure: >1000 HU, SSD >10 cm, lower pole >1 cm (Lower Pole I: 21% vs 91%).
> - 60–90 shocks/min with ramping; good coupling.
> - Use is declining as URS grows; burst wave lithotripsy is coming.

Q3a. Management of Urethral meatal stenosis.

Type: SAQ (10 marks)

Definition and causes

- Narrowing of the external urethral meatus below the normal calibre for age.

Age	Normal calibre
Under 4 months	6–10 Fr
1–12 months	10–12 Fr
1–10 years	12–14 Fr
Over 10 years	14–18 Fr

- **Children:** almost always after **neonatal circumcision**: wet nappies irritate the exposed meatus (ammoniacal meatitis) and the ventral lips fuse into a thin membrane. Rarely congenital with hypospadias.
- **Adults:** LS (lichen sclerosus, formerly BXO [balanitis xerotica obliterans]); catheter or instrument trauma; failed hypospadias repair. Usually the tip of a **fossa navicularis stricture**.

Clinical features

- Thin, forceful stream deflected **upwards**; straining, frequency, incomplete emptying; pain at the start of voiding.

Evaluation

- Watch the child void; calibrate with a bougie à boule. Uroflowmetry. Urethrography only for UTI (urinary tract infection) or suspected proximal lesion.
- Adults: **retrograde urethrogram (RUG) before meatal surgery**, as a proximal stricture often coexists, especially in LS.

Management

Situation	Treatment
Meatitis, early narrowing (child)	Topical 0.05% clobetasol; gentle dilation; nappy hygiene
Symptomatic stenosis (child)	**Ventral meatotomy** (EMLA [eutectic mixture of local anaesthetics] cream in clinic, or general anaesthesia); parents spread the edges with a dilator tip 3 times daily for 7–10 days
Simple adult stenosis	Ventral meatotomy with 3 sutures (apex and each side), mucosa to glans; dilator afterwards
Scarred or recurrent	Dorsal meatotomy (Y-V plasty after excising the scar ridge); effective but poorer cosmesis
Fossa navicularis involved	Urethroplasty: penile skin island flap (Jordan, De Sy), not in LS
LS	Topical steroid; extended meatotomy; **buccal mucosa graft** (never genital skin); perineal urethrostomy for pananterior disease
Unfit or elderly	Self-dilation with clobetasol as lubricant

Complications

- Bleeding, restenosis (commoner in LS), spraying.

Also know: biopsy atypical LS (squamous carcinoma risk up to 8%).

Diagram to draw: ventral meatotomy: crush, incise ventrally, suture mucosa to glans at apex and both sides.

> **Key points to remember:**
>
> - Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
> - Adult: RUG first; often a fossa navicularis stricture.
> - LS: steroid, buccal mucosa, never genital skin.

Q3b. Flexible Ureteroscopy.

Type: SAQ (10 marks)

Covered in part in 2017-Summer Paper 1, Q3b; written in full here.

Introduction

- Flexible ureteroscopy (fURS), or RIRS (retrograde intrarenal surgery), examines and treats the whole upper tract through the natural route.

Instrument

Feature	Details
Tip	About 4.5–7.5 Fr (shaft larger)
Working channel	3.6 Fr (laser fibre, basket)
Deflection	≥270° both ways
Types	**Fibreoptic** (image degrades with fibre breakage); **digital** (chip on tip, better image); **single-use** (no repair or sterilisation cost)

Indications

- **Stones:** renal <2 cm; lower pole <1 cm (or larger as an alternative to PCNL [percutaneous nephrolithotomy]); patients who must continue anticoagulants; failed SWL (shock wave lithotripsy); obesity; bleeding disorders.
- **Others:** diagnosis and laser ablation of low-grade UTUC (upper tract urothelial carcinoma); filling defects, positive cytology; strictures and endopyelotomy; calyceal diverticulum; essential haematuria; foreign bodies.

Technique

1. Sterile urine; single-dose antibiotic.
2. Cystoscopy, **safety guidewire**, retrograde pyelogram.
3. **Ureteral access sheath** (often 12/14 Fr): lowers intrarenal pressure and allows repeated passes, but injures the ureter in nearly half (severe in about 1 in 8).
4. Low-pressure irrigation. **Holmium:YAG (yttrium-aluminium-garnet) or thulium fibre laser:** dusting or fragmenting and basketing.
5. Lower pole stones: move them to an upper calyx with a basket.
6. Inspect the ureter on withdrawal. Omit the stent if the case was uncomplicated with normal anatomy and no ureteric injury.

Outcomes

- Stone-free rate is higher than SWL for renal stones <2 cm but falls with size and lower pole location. Larger stones may need staged procedures.
- Reported rates depend on the imaging used (CT strictest).

Complications

- Stent symptoms (commonest); sepsis about 5%; stricture about 3%; perforation a few per cent; bleeding or subcapsular haematoma <1%; avulsion <1% (nephrostomy, delayed reconstruction); scope damage.

Recent advances

- Suction access sheaths; intrarenal pressure sensors; thulium fibre laser; robotic fURS.

Diagram to draw: flexible scope through an access sheath, deflected into the lower pole calyx.

> **Key points to remember:**
> - 3.6 Fr channel; ≥270° deflection.
> - Access sheath lowers pressure but may injure ureter.
> - First choice when anticoagulants cannot be stopped.

Q3c. Testicular aspiration of sperms and assisted reproduction

Type: SAQ (10 marks)

Introduction

- TESA (testicular sperm aspiration) retrieves sperm from the testis by needle for ICSI (intracytoplasmic sperm injection), which needs only one sperm per oocyte.

Indications

- **OA (obstructive azoospermia)** with normal spermatogenesis: CBAVD (congenital bilateral absence of the vas deferens), unreconstructable obstruction, failed vasectomy reversal, or when epididymal aspiration fails.
- Anejaculation (spinal cord injury) after failed vibratory stimulation or electroejaculation.
- **NOA (non-obstructive azoospermia):** poor yield; micro-TESE preferred.

Technique

1. Local anaesthesia with spermatic cord block.
2. 23 G needle on a high-suction syringe; many passes draw tubules into the needle.
3. Wet examination at once; repeat until sperm are found. Cryopreserve spare sperm.

Comparison of retrieval methods

Technique	Site	Best for	Notes
PESA (percutaneous epididymal sperm aspiration)	Caput epididymis	OA	Few sperm; may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule at 16–25×	OA (best)	Many motile sperm; multiple frozen vials
TESA	Testis, needle	OA	Least invasive; immotile, few sperm
TESE (testicular sperm extraction)	Open biopsy	OA, NOA	Gold standard for diagnosis
Micro-TESE	Larger tubules chosen under microscope	**NOA (best)**	Sperm in about half

Assisted reproduction

- **IUI (intrauterine insemination):** washed sperm into the uterus; mild male factor.
- **IVF (in vitro fertilisation):** oocytes and sperm combined in the laboratory.
- **ICSI:** one sperm injected into each oocyte; essential for surgically retrieved sperm (poor motility).
- Frozen retrieved sperm work as well as fresh, except possibly in spinal cord injury.

Outcomes and concerns

- OA: retrieval almost always succeeds; good pregnancy rates.
- NOA: retrieval depends on the cause (AZFc [azoospermia factor c] deletion 70%, Klinefelter 68%, Sertoli-cell-only 30%, **AZFa/AZFb deletion 0%**). More miscarriages occur.
- ICSI bypasses natural selection: risk of transmitting genetic defects. Offer karyotype, Y-microdeletion and CFTR (cystic fibrosis transmembrane conductance regulator) testing.
- Female burden: ovarian stimulation, multiple pregnancy.
- **TESA complications:** haematoma, pain, vascular injury.

Diagram to draw: needle in testis with syringe suction; flow to ICSI.

Key points to remember:

- TESA: 23 G needle, suction, local anaesthesia; for OA.
- NOA: micro-TESE best; AZFa/AZFb deletion means no sperm.
- All surgically retrieved sperm need ICSI; cryopreserve extras.

Q3d. Management of detrusor underactivity

Type: SAQ (10 marks)

Definition

- **DUA (detrusor underactivity), ICS (International Continence Society) 2002:** "a contraction of reduced strength and/or duration, resulting in prolonged bladder emptying and/or failure to achieve complete bladder emptying within a normal time span." **Acontractile detrusor:** no contraction.
- **UAB (underactive bladder):** symptoms of slow stream, hesitancy and straining, with or without incomplete emptying. Found in 40–48% of men over 70 having urodynamics.

Causes

- **Neurogenic:** cauda equina, disc prolapse, spinal injury, diabetes, multiple system atrophy, early stroke, radical pelvic surgery.
- **Myogenic:** long-standing BOO (bladder outlet obstruction), diabetes, ischaemia, ageing.
- **Drugs:** anticholinergics, opioids. **Functional:** Fowler syndrome, dysfunctional voiding.

Evaluation

- Bladder diary, flow rate, PVR (post-void residual), sacral neurological examination, spinal MRI if indicated, drug review.
- **Pressure-flow study** is the only accepted test. **BCI (bladder contractility index) = Pdet@Qmax (detrusor pressure at maximum flow) + 5 × Qmax:** <100 weak, 100–150 normal, >150 strong. Low Pdet with low Qmax.
- Ambulatory urodynamics if no contraction on standard study (the "bashful bladder").

Management

- **Goals:** protect kidneys, prevent UTI (urinary tract infection) and stones, improve symptoms.

Option	Details
Conservative	Timed and double voiding; treat constipation; stop causative drugs
CISC (clean intermittent self-catheterisation)	**Mainstay** for high PVR; fewer infections than indwelling catheter. Suprapubic catheter if CISC impossible
α-blockers	Lower outlet resistance; modest benefit
Parasympathomimetics (bethanechol, distigmine)	**Weak evidence**, systemic side effects (bronchospasm, rarely cardiac arrest); not routine
SNM (sacral neuromodulation)	For non-obstructive retention, especially Fowler syndrome; needs some detrusor function
Sacral anterior root stimulator (Brindley)	Complete spinal cord injury, with posterior rhizotomy
Intrasphincteric botulinum toxin	Experimental
Outlet surgery (TURP, transurethral resection of the prostate)	Little help in low-PVR men; may free catheter-dependent fit men
Diversion	Mitrofanoff when urethral CISC fails; rarely ileal conduit

- Avoid Credé or Valsalva voiding (reflux risk).

Investigational

- Intravesical electrotherapy, intravesical prostaglandins, TAC-302, latissimus dorsi detrusor myoplasty, muscle-derived stem cells.

Diagram to draw: pressure-flow plot on the Schäfer nomogram with BCI lines at 100 and 150.

> **Key points to remember:**
> - BCI = Pdet@Qmax + 5Qmax; <100 weak.
> - CISC is the mainstay; no effective drug.
> - SNM helps non-obstructive retention; intact sensation predicts response.

Q3e. Retro peritoneal fibrosis medical management

Type: SAQ (10 marks)

Introduction

- RPF (retroperitoneal fibrosis): inflammatory fibrous tissue around the distal aorta (L4–L5) that encases the ureters. Age 40–60; male:female 2–3:1.
- **Idiopathic (about 70%):** chronic periaortitis, often autoimmune. Many cases are **IgG4-related disease** (raised serum IgG4, IgG4-positive plasma cells, storiform fibrosis).
- **Secondary (about 30%):** drugs (methysergide, ergots, β-blockers), malignancy (lymphoma commonest), radiotherapy, asbestos, infection (TB, tuberculosis).

Evaluation before treatment

- Raised ESR (erythrocyte sedimentation rate) and CRP (C-reactive protein), anaemia, renal failure; serum IgG4.
- CT/MRI: mass around aorta and IVC (inferior vena cava), medially deviated ureters; high T2 signal means activity. **PET-CT** is the most sensitive test of activity.
- Biopsy if malignancy is suspected.

Medical management

1. **Relieve obstruction first:** stents or nephrostomy (nephrostomy if uraemic and anuric); manage post-obstructive diuresis.
2. **Stop the culprit drug;** treat any secondary cause.
3. **Steroids are first line:**
    - Prednisone about 60 mg daily (about 1 mg/kg), tapered to 5 mg, total course up to 2 years.
    - About 80% respond: pain and symptoms settle in days, ESR falls, the mass shrinks.
    - Relapse on tapering is 25–50%. Active inflammation predicts response; burnt-out fibrosis responds poorly.
4. **Tamoxifen:** thought to inhibit fibroblasts. **Vaglio RCT (randomised controlled trial; Lancet 2011):** relapse 17% with prednisone vs 50% with tamoxifen, so steroids remain first line.
5. **Steroid-sparing or second line:** mycophenolate mofetil (with steroids), azathioprine, cyclophosphamide, ciclosporin, colchicine; medroxyprogesterone.
6. **Rituximab:** for refractory or IgG4-related disease.

Monitoring

- Symptoms, ESR/CRP, creatinine, ultrasound, periodic CT/MRI or PET-CT. Remove stents when ureters drain.
- Watch steroid side effects (diabetes, osteoporosis).

When medical therapy fails

- About two-thirds clear obstruction with drainage plus drugs; about 15% need **ureterolysis**: bilateral, with biopsies, and the ureters protected by intraperitonealisation or omental wrap (open, laparoscopic or robotic). Autotransplantation is rarely needed.

Diagram to draw: mass around aorta at L4–L5 pulling both ureters medially.

Key points to remember:

- Drain first, then steroids (about 80% respond).
- Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
- Rituximab for IgG4-related or refractory disease; ureterolysis for about 15%.

Q3f. Female Circumcision

Type: SAQ (10 marks)

Assumed to mean female genital mutilation/cutting. Not covered in the reference chapters; answer follows WHO (World Health Organization) definitions.

Definition

- **FGM (female genital mutilation):** all procedures that partly or totally remove the external female genitalia, or otherwise injure them, for non-medical reasons (WHO).
- Over 200 million living women are affected, mainly in Africa, the Middle East and Asia; usually in childhood. Practised in some Indian communities ("khatna").

WHO classification

Type	Description
I (clitoridectomy)	Partial or total removal of the clitoral glans and/or prepuce
II (excision)	Removal of the clitoral glans and labia minora, with or without the labia majora
III (infibulation)	Narrowing of the vaginal opening by cutting and joining the labia into a seal, with or without clitoral excision
IV (other)	Pricking, piercing, incising, scraping, cauterisation

Complications

- **Immediate:** pain, haemorrhage, shock, infection (tetanus), **acute urinary retention**, urethral injury, death.
- **Urological:** slow stream, dysuria, **recurrent UTI (urinary tract infection)**, meatal obstruction under the scar, urethral stenosis, incontinence, **vesicovaginal fistula** after obstructed labour.
- **Gynaecological and sexual:** inclusion cysts, keloid, dyspareunia, infertility.
- **Obstetric:** obstructed labour, perineal tears, postpartum haemorrhage, neonatal death.
- **Psychological:** anxiety, depression, post-traumatic stress disorder.

Evaluation

- Sensitive history; examination to classify type; flow rate, residual urine; cystoscopy if fistula suspected.

Management

- **Deinfibulation (type III):** anterior midline incision of the scar under local or regional anaesthesia to expose the urethral meatus and introitus; raw edges oversewn.
  - **Timing:** ideally before pregnancy, or antenatally, or in labour. Relieves retention, UTI and dyspareunia.
  - **Never reinfibulate** (illegal in the UK).
- Excise cysts; treat stenosis; repair fistula; counselling. Clitoral reconstruction has limited evidence.

Legal and ethical aspects

- WHO and the UN (United Nations) condemn FGM as a human rights violation; WHO opposes its performance by health workers.
- Illegal in many countries (e.g. UK FGM Act 2003, with mandatory reporting in girls under 18). India has no specific law.

Diagram to draw: vulva showing types I–III and the deinfibulation incision line.

Key points to remember:

- WHO types: I clitoridectomy, II excision, III infibulation, IV others.
- Urological: retention, recurrent UTI, stenosis, fistula.
- Deinfibulation for type III; never reinfibulate.

Rapid Revision Sheet – 2019-Summer Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe in detail the management of T1G3 TCC of urinary bladder? (LAQ)

- Re-TURBT in all T1 at 2–6 weeks; residual T1 means progression up to 80%.
- BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
- Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy.
- BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.

Q2. Describe indication, mechanism of stone fragmentation and current status of ESWL? (LAQ)

- Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
- Failure: >1000 HU, SSD >10 cm, lower pole >1 cm (Lower Pole I: 21% vs 91%).
- 60–90 shocks/min with ramping; good coupling.
- Use is declining as URS grows; burst wave lithotripsy is coming.

Q3a. Management of Urethral meatal stenosis. (SAQ)

- Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
- Adult: RUG first; often a fossa navicularis stricture.
- LS: steroid, buccal mucosa, never genital skin.

Q3b. Flexible Ureteroscopy. (SAQ)

- 3.6 Fr channel; ≥270° deflection.
- Access sheath lowers pressure but may injure ureter.
- First choice when anticoagulants cannot be stopped.

Q3c. Testicular aspiration of sperms and assisted reproduction (SAQ)

- TESA: 23 G needle, suction, local anaesthesia; for OA.
- NOA: micro-TESE best; AZFa/AZFb deletion means no sperm.
- All surgically retrieved sperm need ICSI; cryopreserve extras.

Q3d. Management of detrusor underactivity (SAQ)

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
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